

EVOLUTION OF DISLOCATION DENSITIES AND THE CRITICAL CONDITIONS FOR THE PORTEVIN-LE CHÂTELIER EFFECT

L. P. KUBIN¹ and Y. ESTRIN²

¹CNRS-ONERA, LEM, 29, Av. de la Division Leclerc, B.P. 72, 92322 Châtillon Cedex, France
and ²Technical University Hamburg-Harburg, Eissendorfer Strasse 42, 2100 Hamburg 90, F.R.G.

(Received 15 March 1989; in revised form 11 October 1989)

Abstract—A model is proposed for the critical strains associated with the Portevin–Le Châtelier effect (PLC) in terms of the strain dependence of the densities of mobile and forest dislocations. The classical critical condition for the onset of the PLC effect, viz. that of vanishing of the strain rate sensitivity of flow stress under the influence of dynamic strain aging is reexamined. The analysis takes into account the strain dependence of a key quantity: the elementary strain produced when all mobile dislocations perform a successful thermally activated step through the forest obstacles. This elementary strain is estimated by studying a system of coupled differential equations for the evolution of the two densities. Results are obtained in semi-quantitative form and compared with available data. It is shown that the following effects are consistently explained: the occurrence of critical strains for the onset and termination of jerky flow, occasional observation of two PLC regimes within the same deformation curve, the behaviour of the critical strains at high strain rates and low temperatures and, possibly, the particular behaviour exhibited by some alloys at low strain rates and high temperatures. Consequences for the “friction” and “forest” models of dynamic strain aging are discussed.

Résumé—Un modèle tenant compte de la variation avec la déformation des densités de dislocations mobiles et de la forêt est proposé pour les déformations critiques associées à l'effet Portevin–Le Châtelier (PLC). La condition critique classique pour l'application de l'effet PLC suivant laquelle la sensibilité à la vitesse de la contrainte d'écoulement s'annule sous l'effet du vieillissement dynamique est réexaminée. L'analyse est effectuée en fonction d'une quantité essentielle: la déformation élémentaire obtenue lorsque toutes les dislocations mobiles effectuent un mouvement thermiquement activé à travers les obstacles de la forêt. Cette déformation élémentaire et sa dépendance en déformation sont calculées au moyen d'un système d'équations différentielles couplées pour l'évolution des deux densités. Les résultats obtenus sous forme semi-quantitative sont comparés avec les données de la littérature. Les effets suivants peuvent alors être expliqués: l'existence de déformations critiques pour l'apparition et la disparition de la déformation instable, l'observation occasionnelle de deux régimes successifs d'effet PLC sur une même courbe de déformation, l'évolution des déformations critiques à forte vitesse de déformation et à basse température, peut-être également le comportement particulier de certains alliages à faible vitesse et à haute température. Les conséquences concernant les modèles de “friction” et de “forêt” du vieillissement dynamique sont discutées.

Zusammenfassung—Es wird ein Modell zur Beschreibung der kritischen Dehnungen vorgestellt, die für den Portevin–Le Châtelier-Effekt (PLC) charakteristisch sind. Dem Modell liegt die Dehnungsabhängigkeit der Dichten von beweglichen und Waldversetzungen zugrunde. Die klassische kritische Bedingung für das Einsetzen des PLC-Effektes, die darin besteht, daß die Geschwindigkeitsempfindlichkeit der Fließspannung als Folge dynamischer Reckalterung gleich Null wird, wird überprüft. Hierbei wird der Analyse der Dehnungsabhängigkeit der Elementardehnung eine Schlüsselrolle beigemessen. Diese Größe wird als Dehnungsinkrement definiert, der durch einen thermisch aktivierten Schritt aller beweglichen Versetzungen im Hindernisfeld der Waldversetzungen erzeugt wird. Diese Elementardehnung wird abgeschätzt, indem ein Differentialgleichungssystem für die beiden gekoppelten Versetzungsdichten untersucht wird. Die Ergebnisse sind in halbquantitativer Form präsentiert. Sie werden mit vorhandenen Meßdaten verglichen. Es wird gezeigt, daß folgende Effekte schlüssig erklärt werden können: das Auftreten der kritischen Dehnungen, bei denen diskontinuierliches plastisches Fließen einsetzt bzw. aufhört; vereinzelte Beobachtungen des Auftretens zweier PLC-Bereiche innerhalb einer und derselben Verformungskurve; das Verhalten der kritischen Dehnungen bei hohen Dehngeschwindigkeiten und niedrigen Temperaturen und—in bestimmten Legierungen—bei niedrigen Dehngeschwindigkeiten und hohen Temperaturen. Die Auswirkungen des Modells für das “Reibungs”- bzw. “Waldmodell” der dynamischen Reckalterung werden diskutiert.

1. INTRODUCTION

In many alloys, jerky flow associated with the Portevin–Le Châtelier (PLC) effect commences after a finite amount of strain, ϵ_c . Termination of jerky flow at a larger value of strain, ϵ'_c , has also been

observed in some cases. In the most commonly accepted models jerky flow (serrated yielding) is related to the dynamic interaction between diffusing solute atoms and mobile dislocations, i.e. to dynamic strain aging (DSA). DSA induces a negative contri-

bution to the total strain rate sensitivity (SRS) of flow stress and the onset of the PLC effect occurs when the (strain dependent) SRS turns negative [1, 2]†.

To date, there is no consensus about the mechanisms responsible for this strain dependence of the SRS (cf. e.g. Ref. [4]). Three main possible causes have been discussed: (i) enhancement of solute atom diffusivity by deformation-induced vacancies, as initially proposed by Cottrell [5], (ii) increase in mobile dislocation density, and (iii) redistribution of solute atoms in the distorted regions of the dislocation cores [6–8].

According to van den Beukel [9], both mechanisms (i) and (ii) contribute to the value of ϵ_c in substitutional alloys, while only mechanism (ii) should be operative in interstitial alloys, where a critical strain has also been recorded [10–12]. Numerical estimates based on this hypothesis make use of phenomenological expressions for the evolution of the vacancy concentration and of the mobile dislocation density, which are assumed to increase as some power of the plastic strain. In such models the critical strain ϵ_c is obtained from the condition that the SRS vanishes. A connection is then established between DSA mechanisms and the critical strain from which the strain rate and temperature dependence of the latter can be derived in principle [9].

In the present study the “dislocation model” is investigated in some detail. It is shown that by refining the strain dependence of the densities of mobile and forest dislocations it is possible to obtain a rather realistic description of the critical strains ϵ_c and ϵ'_c , without necessarily assuming that diffusion is vacancy-assisted. It can be viewed as an extension of the model (ii) based on the variation of the mobile dislocation density. The SRS depends on these two densities through a parameter Ω which is the strain increment produced when all the mobile dislocations accomplish a successful thermally activated event (cf. Ref. [2], where this quantity is labelled ϵ_0). This definition does not imply, of course, that all dislocations perform this simultaneously, in a coherent way. Since the obstacles to be overcome during DSA are localized at intersections of mobile dislocations with forest dislocations, Ω now depends on both dislocation densities. Their evolution leads to a characteristic strain profile of Ω .

In Section 2, the critical strains are defined from the condition of vanishing of the SRS. This quantity is estimated along the lines of the classic models [1, 2]. Because of a particular strain profile of Ω , the critical condition of vanishing of the SRS may be met several times (up to four times) along a deformation curve. The strain dependence of Ω and its sensitivity to

the choice of the elementary dislocation mechanisms are examined in more detail in the Appendix. For this purpose we make use of a differential system analogous to the one proposed in [13] to represent the coupled evolution with strain of the densities of mobile and forest dislocations: ρ_m and ρ_f respectively. Various possible experimental situations are examined in Section 3 with reference to the data available in the literature. Finally, it is suggested that current debates about the strain dependence of dynamic strain aging [4] could benefit from a more refined description of the elementary strain.

2. DISLOCATION MODEL OF THE CRITICAL STRAINS

2.1. Strain rate sensitivity during dynamic strain aging

As in former studies [1, 2, 15], we assume that the SRS can be expressed as a sum of two additive contributions. One, S_0 , stems from the activation of dislocation motion in the absence of DSA. The contribution from DSA, S_{aging} , is estimated as follows: during their waiting times at localized forest obstacles, dislocations get pinned by solute atoms which diffuse to them. In consequence, the forest obstacles become “harder” with increasing arrest time, t . This hardening is represented by a function $f(t, T)$ which is assumed to be proportional to the solute concentration around the obstacle (here T is the absolute temperature). The solute concentration is determined by specifying the solute diffusion mechanism which is obviously temperature dependent. When a successful activation event takes place, the aging time is equal to the waiting time, t_w , and the obstacle strength has increased by $f(t_w)$ owing to DSA.

The average spacing between localized obstacles is of the order of $\rho_f^{-1/2}$ and considering the free-flight time between obstacles as negligible, the average velocity of the mobile dislocations is proportional to $\rho_f^{-1/2}/t_w$. With the accuracy of a geometrical and a statistical factor, the plastic strain rate, $\dot{\epsilon}$, is then given by

$$\dot{\epsilon} = b\rho_m\rho_f^{-1/2}/t_w = \Omega/t_w. \quad (1)$$

(Here b is the magnitude of the Burgers vector.) This defines the elementary incremental strain $\Omega = \Omega(\epsilon)$, a strain dependent quantity which corresponds to the deformation obtained when all mobile dislocations accomplish a successful activation event. With the help of equation (1), we can now express $f(t_w)$ as $f(\Omega, \dot{\epsilon})$, or alternatively $f(\epsilon, \dot{\epsilon})$, and estimate the contribution of DSA to the total SRS

$$S_{\text{aging}}(\epsilon) = (\partial f / \partial \ln \dot{\epsilon})_{\epsilon} = -df/d \ln t_w. \quad (2)$$

Finally, the total SRS is

$$S(\epsilon) = S_0 + S_{\text{aging}}(\epsilon) = S_0 - df/d \ln [\Omega(\epsilon)/\dot{\epsilon}]. \quad (3)$$

Within some differences in labelling, and of course in the way the strain dependence is introduced, equation (3) recovers an expression proposed by

†Actually, the critical strain ϵ_c is sometimes found to be larger than the strain at which the SRS vanishes. This delay in the onset of serrated yielding, which will not be considered here, has been recently discussed by McCormick [3].

van den Beukel [1] and further used by other authors [2, 15]. The SRS in the absence of DSA, S_0 , is generally a variable quantity. Thus, in materials exhibiting the so-called Cottrell-Stokes behaviour, S_0 is proportional to stress and hence depends on ϵ . This is typically the case for f.c.c. metals. For the sake of simplicity, S_0 will be considered constant, however.

2.2. The critical strains

A commonly met experimental situation is as follows: the SRS is positive at small strains, but decreases as plastic flow proceeds. Serrated yielding commences after an incubation strain, when the critical condition $S(\epsilon) = 0$ is reached. It may disappear at a later stage, provided that S turns positive again. If S is negative from the outset, $S(0) < 0$, there is no incubation strain. To estimate the critical conditions we make use of the function $f(t_w, T)$ introduced by Louat [16]

$$f = f_0 \{1 - \exp[-(t_w/\tau_0)^{2/3}]\}. \quad (4)$$

Here f_0 is the maximum increase in stress which can be produced by the aging mechanism and τ_0 is the relaxation time associated with diffusion which depends on the binding energy between a solute atom and a dislocation, on the diffusion coefficient of solute atoms and on solute concentration [17]. The time dependence expressed by equation (4) recovers the $t^{2/3}$ law at small times that follows from the classic theory [5] for dislocation-solute interactions via the size effect. It also describes the saturation behaviour in the limit of large time and provides a convenient interpolation between the two limits (cf. e.g. Ref. [15]). This particular choice of the aging function captures the basic features of the aging process, but does not rule out other types of interactions or diffusional processes. Expressing the condition that $S = 0$ in equation (3) and inserting f from equation (4), we obtain

$$X \exp(-X) = \alpha \quad (5)$$

where $\alpha = (3/2)(S_0/f_0)$ and $X = (t_w/\tau_0)^{2/3} = (\Omega/Z)^{2/3}$, with $Z = \dot{\epsilon}\tau_0$. In the dimensionless quantity X , the effect of the evolution of the dislocation structure is incorporated through the elementary incremental strain Ω . The effect of strain rate and temperature enters through the quantity Z which bears some similarity with the Zener-Hollomon parameter often used in studies on creep. (Note that the relaxation time τ_0 associated with solute diffusion to dislocations is given by an Arrhenius-type formula). As shown in Fig. 1, $X \exp(-X)$ is a bell-shaped function which has a maximum at $X = 1$, i.e. when $t_w = \tau_0$. Depending on the value of the value of α , equation (5) defines two solutions, X_1 and X_2 ($X_2 > X_1$), or one tangent solution (when $\alpha = 1/e$). There is no solution when $\alpha > 1/e$. Given the strain rate and the temperature, this defines at most two critical values of the incremental strain,

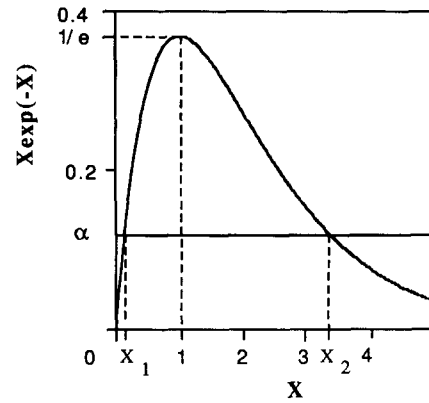


Fig. 1. Determination of the critical values of the reduced quantity $X = (\Omega/Z)^{2/3}$.

($\Omega_{1,2} = Z(X_{1,2})^{3/2}$), the SRS being negative within the Ω interval (Ω_1, Ω_2).

2.3. The strain profile of the elementary incremental strain

At small strains the mobile density should increase with strain faster than the forest density [13], and, therefore, faster than $\rho_f^{1/2}$. We then expect that $\Omega = b\rho_m\rho_f^{-1/2}$ first increases with strain, starting from an initial value Ω_0 . At large strains, the density of forest dislocations increases faster than that of mobile dislocations which is close to saturation [13]. Ω decreases then as $b\rho_{ms}\rho_f^{-1/2}$ asymptotically approaching the value $\Omega_s = b\rho_{ms}\rho_{fs}^{-1/2}$. Here ρ_{ms} and ρ_{fs} denote the saturation values of the mobile and forest dislocation density respectively. By inserting reasonable values for the initial and saturation densities, one can easily prove that typically $\Omega_0 < \Omega_s$, cf. Appendix and Fig. 3. The behaviour of Ω at large strains can also be rationalized by observing that $\rho_f^{1/2}$ is proportional to the flow stress σ . At large strains Ω will decrease nearly as the inverse of σ and eventually saturate to a value proportional to $1/\sigma_s$ where σ_s is the saturation value of stress. Thus, $\Omega(\epsilon)$ is expected to pass through a maximum Ω_m (cf. Fig. 3) at an intermediate strain $\epsilon = \epsilon_m$.

In the Appendix, the assessed shape of $\Omega(\epsilon)$ is substantiated through numerical arguments. A differential system, analogous to the one investigated in a previous work [13] describes the evolution with strain of two coupled quantities: the mobile and forest densities. By attributing reasonable numerical values to the material parameters involved, typical solutions $\rho_m(\epsilon)$, $\rho_f(\epsilon)$ and $\Omega(\epsilon)$ can be computed, as illustrated by Figs. 2 and 3, curve (a). These results will be used in what follows for order-of-magnitude estimates.

The numerical checks reported in the Appendix demonstrate that the Ω vs ϵ profile exhibits a rather flat maximum. The position and height of this maximum are, however, rather sensitive to the values of the material constants determining the elementary dislocation mechanisms and possibly also to strain

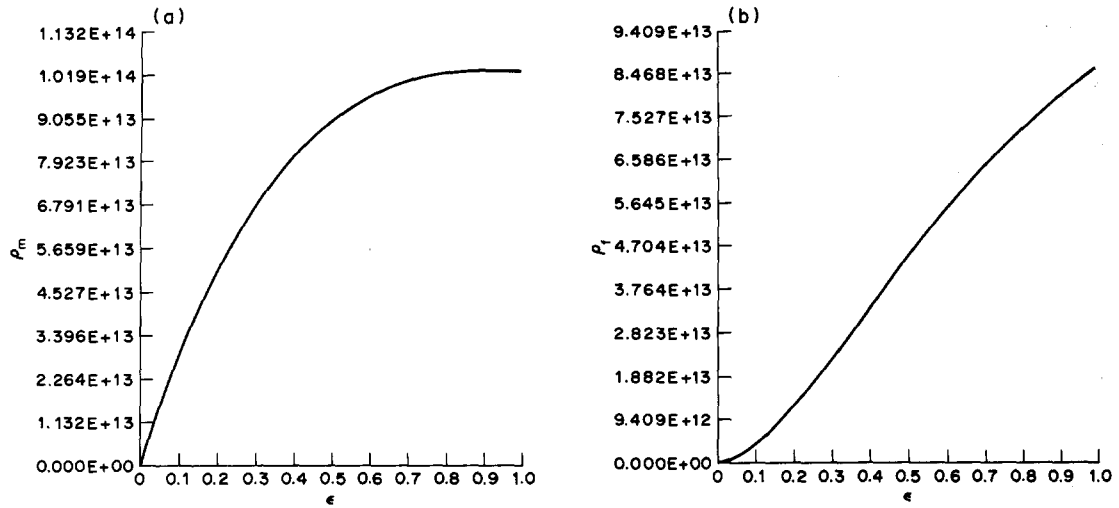


Fig. 2. Typical strain dependences of the mobile dislocation density ρ_m (a) and the forest dislocation density ρ_f (b) for the basic set of parameter values (see Appendix); $Z = 0.05$.

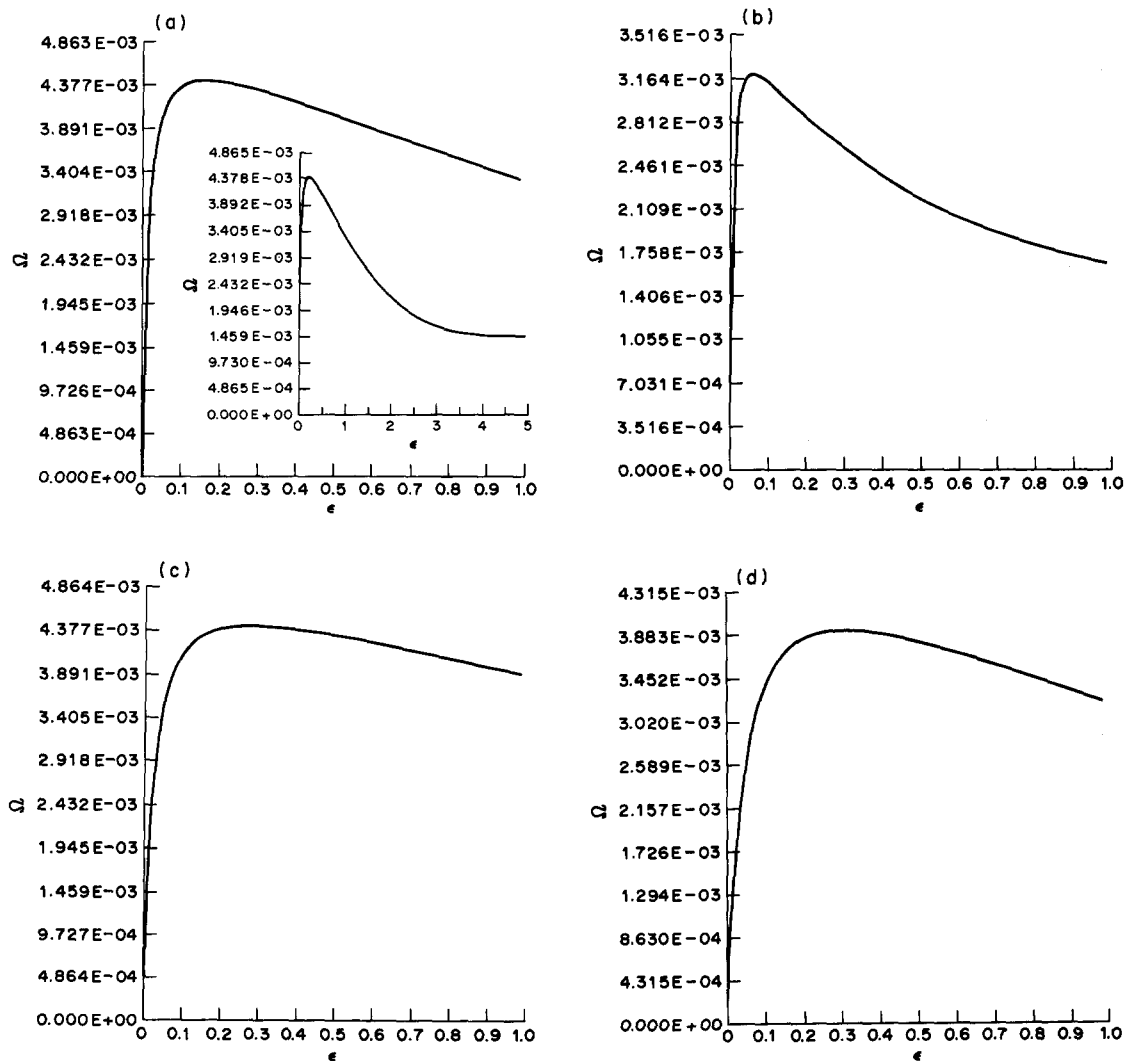


Fig. 3. Strain dependence of the elementary strain: (a) as determined from the basic set of parameter values defined in the Appendix. (A large scale view is shown in the insert); (b) same as (a), but with $\lambda = 1$; (c) same as (a), but with $\epsilon_f = 0.5$; (d) same as (a), but with $\rho_0 = 10^{12} \text{ m}^{-2}$.

rate and temperature through e.g. thermally activated recovery. This can be visualized from curves (b) to (d) of Fig. 3 which were obtained by exploring the possible domain of variation of several material constants. The initial value of ρ_m is only important at very small strains (below 0.1%, say), so that the Ω vs ϵ profile is not sensitive to this quantity in the strain range of interest.

2.4. The range of existence of jerky flow

The critical strains are obtained by searching for the values of strain which satisfy the condition $\Omega(\epsilon) = \Omega_{1,2}$. In the particular case schematically drawn in Fig. 4(a), there are two solutions, ϵ_c and ϵ_c^* . (Henceforth, intersections with the descending branch of the Ω vs ϵ profile are denoted by asterisked symbols, viz. ϵ_c^* and $\epsilon_c'^*$.) For ϵ within the interval $(\epsilon_c, \epsilon_c^*)$, the SRS is negative as shown in Fig. 4(b), and jerky flow should occur. More generally, the expected number of solutions of the equation $S = 0$ ranges from zero to four. Possible situations are

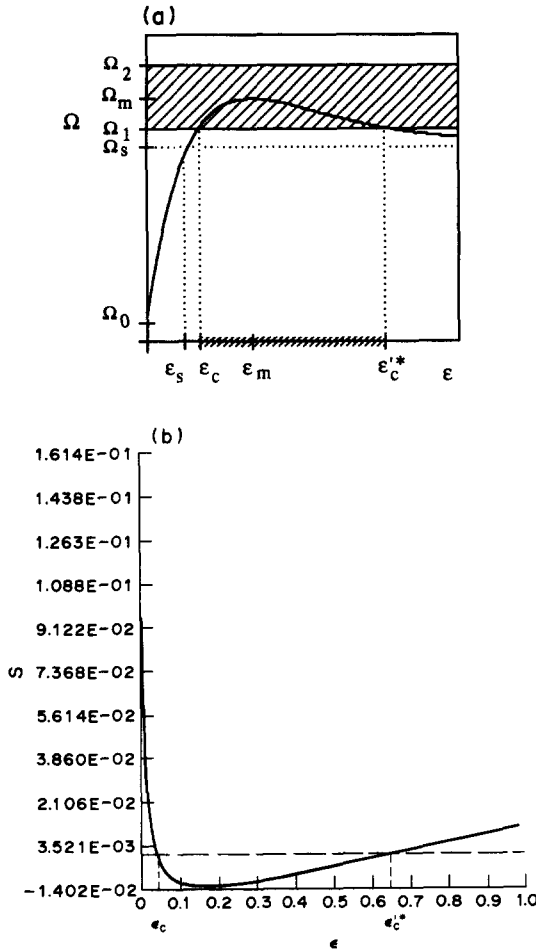


Fig. 4. Determination of the critical strains for the onset (ϵ_c) and termination (ϵ_c^*) of jerky flow. (a) From the strain dependence of the elementary strain (schematic). (b) From the strain dependence of the nondimensional SRS, $s = (3/2)(S/f_0)$. (Basic set of parameter values; $Z = 0.05$.) The horizontal dashed line indicates zero SRS.

investigated and discussed in more detail in the next section. Here we only deal with the conditions for the occurrence of serrations on a deformation curve, i.e. for the existence of at least one solution.

When the test temperature and the applied strain rate are changed, the Ω vs ϵ profile and the critical values $\Omega_{1,2}$ are modified. Since we consider here a somewhat idealized situation, representing the typical behaviour of an undefined model alloy, we assume that, in a first approximation, the main changes arise from the factor Z involved in the dimensionless quantity $X = (\Omega/Z)^{2/3}$. The relaxation time τ_0 associated with solute diffusion is proportional to the inverse of the diffusion coefficient of solute atoms. With regard to our analysis, an increase in temperature is therefore equivalent to a decrease in strain rate. Below we focus on strain rate dependences, from which temperature dependences can be inferred easily.

To obtain at least one critical strain, two conditions must be fulfilled simultaneously:

(i) Equation (5) must have at least one solution, which is realized for $\alpha \leq 1/e$, i.e. the maximum strengthening f_0 produced by DSA must be larger than a threshold value (typically a few S_0).

(ii) In the graph of Fig. 4(a), the band bounded by the horizontal lines $\Omega = \Omega_1$, $\Omega = \Omega_2$, within which the SRS is negative, must intersect the Ω vs ϵ profile. The threshold conditions are $\Omega_2 = \Omega_0$, when the band is located below the elementary strain profile, i.e. at low strain rates and/or high temperatures, or $\Omega_1 = \Omega_m$ when the band is located above the elementary strain profile, i.e. at high strain rates and/or low temperatures. Expressing these conditions in terms of strain rates (cf. Section 2.1), we obtain a bounded range of strain rates at a given temperature (and of temperatures at a given strain rate) within which jerky flow can occur

$$\Omega_0/(\tau_0 X_2^{3/2}) < \dot{\epsilon} < \Omega_m/(\tau_0 X_1^{3/2}). \quad (6)$$

If these conditions are not fulfilled, DSA results in just a decrease of the macroscopic SRS which, however, remains positive so that no instabilities can be obtained.

To illustrate the ideas, we estimate this range of strain rates numerically, by adopting parameter values drawn from a previous study on Al-5% Mg alloys around room temperature [15]: $S_0 = 3$ MPa, $f_0 = 30$ MPa, from which we obtain $\alpha = 0.15$, $X_1 = 0.18$ and $X_2 = 3$. At or around room temperature, τ_0 is typically of the order of 1s [15], within one order of magnitude or more, however. From the numerical estimates of the dislocation densities given in the Appendix [cf. also Fig. 3, curve (a)], we have $\Omega_0 = 9.5 \times 10^{-6}$ and $\Omega_m = 4.4 \times 10^{-3} \approx 3\Omega_0$. Equation (6) yields: $1.8 \times 10^{-6} \text{ s}^{-1} < \dot{\epsilon} < 5.8 \times 10^{-2} \text{ s}^{-1}$, which is quite representative of the range of existence of the PLC effect. Furthermore we see that, in agreement with experiment, these values may be modified by

several orders of magnitude if the material or experimental conditions are changed.

3. THE CRITICAL STRAINS

3.1. Geometrical construction for critical strain determination

In this section we are concerned with the determination of the number and properties of the possible solutions for the condition giving the critical strains, $\Omega_{1,2} = Z(X_{1,2})^{3/2} = \Omega(\epsilon)$. When the strain rate increases, or when temperature decreases, several "scenarios" are possible since (i) the function $\Omega(\epsilon)$ is bivalued between Ω_s and Ω_m , and (ii) we have to consider the intersection of the Ω vs ϵ curve with a strip of negative SRS, whose width, $\Delta\Omega = \Omega_2 - \Omega_1 = Z(X_2^{3/2} - X_1^{3/2})$, increases with increasing strain rate (or decreasing temperatures).

The arguments which follow are quite general since they are based on the global shape of the elementary strain profile. We expect that they reproduce all the experimental complexity associated with the occurrence of critical strains in various materials and in various experimental situations. The only assumption made here, certainly a rather important one, is that the elementary strain profile is not too severely distorted when strain rate or temperature are changed. The procedure of determining the critical strains thus consists in tracing the displacement with increasing strain rate of the strip confined between the straight lines $\Omega = \Omega_1$ and $\Omega = \Omega_2$ and finding the intercepts of these lines with the Ω vs ϵ profile. As an example, we first discuss in some detail the determination of one possible sequence, labelled scenario 1, before listing all other possible scenarios.

When the strain rate increases, the first occurrence of jerky flow is recorded when Ω_2 crosses the horizontal line $\Omega = \Omega_0$ (cf. Fig. 5). When Ω_2 moves up to Ω_s , two situations are possible depending on whether Ω_1 is smaller or larger than Ω_0 when $\Omega_2 = \Omega_s$. Assuming that $\Omega_1 < \Omega_0$ for strain rates such that $\Omega_0 < \Omega_2 < \Omega_s$, the critical strain for the onset of serrations, ϵ_c , is zero. There is a critical strain for the disappearance of serrations, ϵ'_c , which increases from 0 up to a value ϵ_s such that $\Omega(\epsilon_s) = \Omega_s$ [cf. Fig. 5(a)]. Further, when Ω_2 is shifted upwards up to Ω_m , another branching occurs depending on whether Ω_1 is smaller or larger than Ω_0 when $\Omega_2 = \Omega_m$. Again, we assume that the former condition is realized. Then, when Ω_2 increases from Ω_s up to Ω_m , the first critical strain ϵ_c remains zero while the second one moves up from ϵ_s to ϵ_m , the strain at which $\Omega(\epsilon)$ is a maximum. The strain range within which jerky flow occurs thus increases, but simultaneously a third critical strain ϵ_c^* appears [Fig. 5(b)]. On further increasing the strain rate, this quantity decreases to ϵ_m . Thus, a second range of jerky flow appears at large strains. The two ranges merge at the strain ϵ_m when $\Omega_2 = \Omega_m$. When the strain rate is further increased, jerky flow is observed all along the deformation curve, until the condition

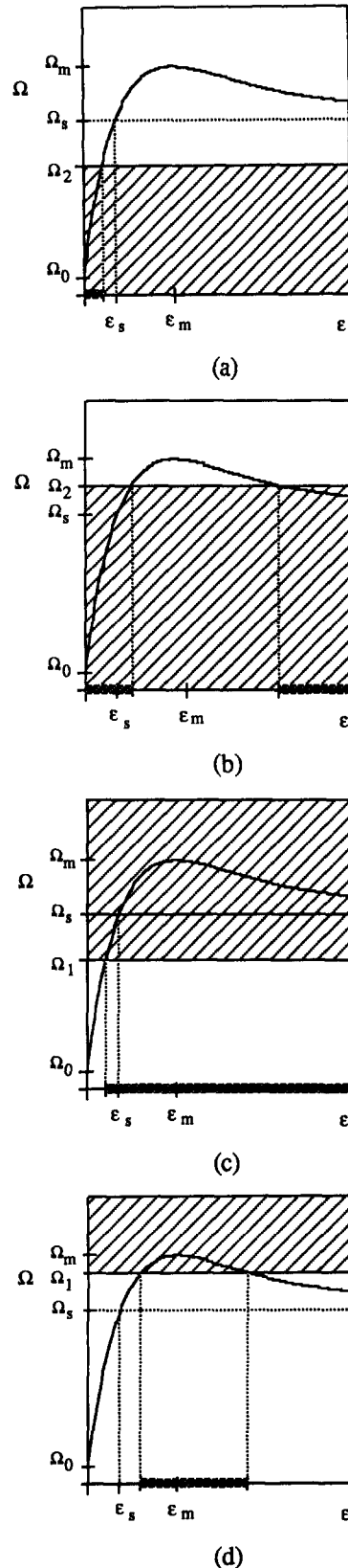


Fig. 5. The geometrical construction of the critical strains with increasing strain rates or decreasing temperatures, according to the particular sequence labelled "Scenario 1".

$\Omega_0 \leq \Omega_1$ is fulfilled. Then, the first critical strain ϵ_c starts increasing [Fig. 5(c)]. From $\Omega_1 = \Omega_s$ onwards an additional critical strain appears which decreases from infinity to ϵ_m . The range of jerky flow thus shrinks until it disappears at the strain ϵ_m when $\Omega_1 = \Omega_m$ [Fig. 5(d)].

This particular sequence of events is summarized in Fig. 6(1). As seen in the preceding paragraph, two conditions must be met for this sequence to occur. One of them can be expressed as follows. When $\dot{\epsilon}\tau_0 X_2^{3/2} = \Omega_m$, the inequality $\dot{\epsilon}\tau_0 X_1^{3/2} < \Omega_0$ has to be fulfilled simultaneously. By eliminating $Z = \dot{\epsilon}\tau_0$ we obtain: $X_1/X_2 < (\Omega_0/\Omega_m)^{2/3}$. This is the condition for the occurrence of scenario 1 since it can be checked easily that the other condition is less stringent. From equation (5) (cf. also Fig. 1) it can be seen that, actually, this condition expresses that the quantity $\alpha = 3S_0/2f_0$ must be smaller than a certain value, implying a sufficiently large strengthening effect of solutes and a small strain rate sensitivity in the absence of DSA.

3.2. Possible scenarios

The conditions for the occurrence of each particular scenario depend on the value of X_1/X_2 with

respect to three quantities: $(\Omega_0/\Omega_m)^{2/3}$, $(\Omega_0/\Omega_s)^{2/3}$ and $(\Omega_s/\Omega_m)^{2/3}$. Considering the graphs of Fig. 4, it seems likely that, in most cases, we have $(\Omega_s/\Omega_m) > (\Omega_0/\Omega_s)$, so that the total number of possible cases reduces to four. They are graphically described in Fig. 6: Fig. 6(1) represents scenario 1 discussed above. The conditions for the other scenarios to realize are as follows $(\Omega_0/\Omega_m)^{2/3} < X_1/X_2 < (\Omega_0/\Omega_s)^{2/3}$ [Fig. 6(2)], $(\Omega_0/\Omega_s)^{2/3} < X_1/X_2 < (\Omega_s/\Omega_m)^{2/3}$ [Fig. 6(3)] and $X_1/X_2 > (\Omega_s/\Omega_m)^{2/3}$ [Fig. 6(4)].

There are several reasons for which it could be experimentally difficult to observe these sequences in full detail. The value of ϵ_s may be too small to be detectable, cf. Fig. 3, where $\epsilon_s \approx 10^{-3}$ for curve (a), but can attain values of the order of 10^{-2} . Because the maximum of the elementary strain profile can be rather flat (cf. Fig. 3), the intersections with the descending branch of $\Omega(\epsilon)$ may very often be pushed to very large strains, in excess of the rupture strain. In such a case, the second range of existence of jerky flow would remain unnoticed. Finally, the shape of the elementary strain profile can be deformed through changes in dislocation mechanisms, particularly those which govern the value of the flow stress at saturation, when strain rate and temperature are

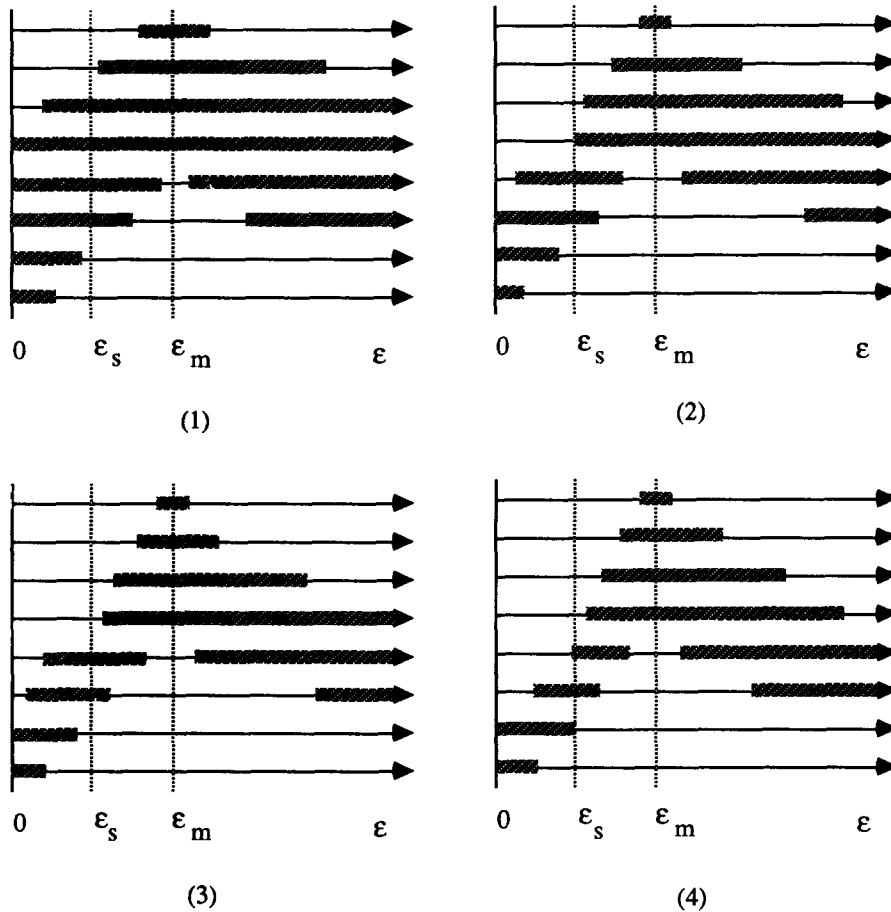


Fig. 6. The four most likely possible sequences of diagrams for the strain rate dependence of the critical strains. Strain rates increase (or temperatures decrease) from bottom to top. The hatched parts indicate the ranges of negative SRS.

varied. Although the maximum should not disappear altogether, this could introduce new variants of the possible scenarios. Thus, the most typical behaviour could be described as follows. When the strain rate increases, a small domain of jerky flow appears at the beginning of the deformation curve and the critical strain increases from zero to a maximum value ϵ_m whose magnitude ranges from a few percent to typically 0.2. The extension of this domain increases and it may be observed to merge with a second domain of jerky flow which has been initiated at large strains. In a certain range of strain rates, jerky flow is observed along almost the entire deformation curve and at a later stage, the domain of jerky flow shrinks and vanishes around a strain equal to ϵ_m .

Numerical estimates performed with the help of the elementary strain profile of Fig. 3(a) yield the following domains of existence for each sequence, in terms of the nondimensional quantity α which measures the strengthening effect of solute atoms: $\alpha < 0.065$ for (1), $0.065 < \alpha < 0.1$ for (2), $0.1 < \alpha < 0.344$ for (3) and $0.344 < \alpha < 1/e$ for (4), i.e. a fairly small domain in this last case. The strain rate values delimitating the domain of jerky flow at room temperature are typically of the order of 10^{-6} and 10^{-2} s^{-1} . As already mentioned, these are only order of magnitude estimates.

Finally, Fig. 7 illustrates in a double logarithmic plot the strain rate dependence of ϵ_c . It turns out to be fairly linear except for a part in proximity to the maximum value obtainable for a given set of parameter values. Numerical values of the DSA characteristics used in this calculation are those defined in Section 2.3, and the elementary strain profile is that of Fig. 3(a).

4. COMPARISON WITH EXPERIMENTAL DATA. DISCUSSION

4.1. The critical strains: general aspects

In the present model, we made use of a heuristic description of DSA to emphasize the fact that the strengthening effect of solute atoms saturates for large waiting times, i.e. at small strain rates and elevated temperatures. This should be true irrespective of the nature of the diffusional process involved (e.g. bulk or pipe diffusion). The model yields a finite domain of negative SRS, which is a prerequisite of the PLC effect, and the doubling of the number of possible critical strains. A few simplifying assumptions were made which could be relaxed in more detailed studies. The increase in flow stress due to aging was considered stress-independent, although this is known not to be always true [18]. In addition, the strain rate sensitivity in the absence of DSA, S_0 , was assumed to be a constant while it follows from the Cottrell-Stokes law that in f.c.c. solid solutions it should linearly increase with stress.

The fact that forest dislocations act as obstacles to the motion of mobile dislocations is another basic

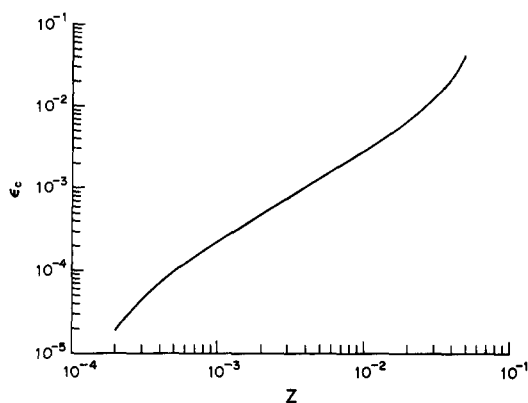


Fig. 7. Numerical evaluation of the strain rate dependence of the first critical strain, ϵ_c , using the elementary strain profiles of Fig. 3(a).

statement of the present model. The elementary strain profile which is a key quantity in this approach was investigated by considering the evolution of the two interrelated densities. The general arguments presented in Section 2 as well as numerical checks indicate that the strain profile of the elementary strain generally exhibits a maximum. As a consequence, up to four successive critical strains can be defined, although we do not expect them to be always accessible to experiment for reasons which have already been explained in the preceding section. Within the present approach one can distinguish between two types of critical strains. Those obtained along the ascending branch of the elementary strain profile, ϵ_c and ϵ'_c , increase with increasing strain rates and decreasing temperatures. They can be said to behave in a thermally activated or "normal" manner. On the contrary, the intersections obtained along the descending branch, ϵ_c^* and ϵ'^*_c , result from dislocation storage effects and behave in an "anti-thermally activated" or inverse manner (cf. Fig. 6).

4.2. The first critical strain: "normal behaviour"

There is a considerable bulk of experimental data on the first critical strain, ϵ_c , which will not be reviewed in detail here. Its "normal" dependence on strain rate or temperature is well-documented for many alloy systems. It has been characterized either over the whole domain of occurrence of jerky flow or, notably in Al-based alloys, only at high strain rates and low temperatures. As shown in Fig. 7, the strain rate dependence of the critical strain ϵ_c for $\epsilon < \epsilon_m$ can be approximated by a power law: a feature that motivated a number of investigators to derive an activation energy commonly identified with that for the diffusional process involved in DSA. Such determination is usually performed assuming conditions far from saturation, i.e. waiting times small compared to the relaxation time for solute diffusion, τ_0 . This may become problematic at low strain rates and high temperatures and may yield non-physical values when applied to other critical strains.

In the example of Fig. 7 the value of the slope, which is close to 1, stems from the particular form of the creation term for the mobile density (cf. Appendix) and from its predominance at small strains. Other types of multiplication mechanisms, such as those discussed in Ref. [14], may lead to different values of the slope of ϵ_c vs $\ln \dot{\epsilon}$ diagrams.

4.3. The first critical strain: inverse behaviour

The "inverse" behaviour has been repeatedly observed in Al-based substitutional solid solutions for low strain rate tests conducted at room temperature and above, cf. [19–22] and earlier references quoted therein. There are similar reports on several plain carbon steels [23, 24], also in a domain of low strain rates and high temperatures. In such cases the double logarithmic plots of critical strain vs strain rate exhibit a "normal" increasing branch and a descending branch of inverse strain rate dependence separated by a minimum (or a portion of zero critical strain). It seems established, by studying the influence of various pre-treatments and the observations of preferential occurrence of a particular type of serrations ("unlocking" serrations) that this anomalous dependence is connected with precipitation before and during the deformation test [21, 22]. In particular, the formation of G.P. zones appears to be quite effective in delaying the onset of serrated flow. The mechanisms by which the critical strain is affected are not understood, however. Earlier interpretations involved an assumption that the mobile dislocation density should decrease with increasing strain [20, 25]. This is perhaps unlikely but would make sense if we consider that, instead, the inverse critical strain corresponds to an intersection of the horizontal line $\Omega = \Omega_c$ with the decreasing branch of $\Omega(\epsilon)$. The following tentative explanation could be proposed here. Precipitation modifies the concentration of solute atoms dissolved in the matrix, but it also may substantially modify the profile of the elementary strain, e.g. through changes in the multiplication mechanism, the rate of dislocation storage, or the degree of homogeneity of plastic flow. This may result in a shift of the maximum in Ω to such small strains that the first domain of jerky flow would become barely noticeable [cf. Fig. 4(b)]. It is the critical strain ϵ_c^* that will be recorded which quantity exhibits an inverse behaviour with strain rate or temperature.

4.4. Other critical strains

Taking into account the sequences depicted in Fig. 6, as well as possible deformations of the profile of elementary strain and the occurrence of inverse critical strains, virtually any type of experimental behaviour could be explained, albeit with various degree of confidence. For this reason, we will not discuss occasional reports on the disappearance of serrated flow in commercial or highly concentrated alloys [7, 18, 26]. The data of Munz and Macherauch

[26] are particularly hard to be understood within the present frame. In a Cu–30 wt%Zn alloy these authors recorded a first critical strain with an inverse behaviour and a second critical strain with a normal behaviour for conventional strain rates and temperatures around 580 K. There are, nevertheless, at least two detailed observations [20, 27] in low concentration binary alloys which confirm the expectations of the present model.

Figure 8, reproduced from Ref. [20], shows the stress–strain curves for an Al–0.5% Mg alloy for a fixed strain rate of $3.33 \times 10^{-4} \text{ s}^{-1}$ and temperatures between 144 and 160°C. In this alloy the Mg concentration is below the solubility limit and the first critical strain exhibits a normal behaviour. The critical strain for the disappearance of jerky flow decreases with increasing temperature, up to the limit of the domain of existence of the PLC effect at 170°C. In parallel, the first critical strain attains vanishing values. This type of behaviour is, indeed, common to all the schemes depicted in Fig. 6. A more striking observation by R  uchle *et al.* [27] on a Cu–3.3 at.% Sn alloy is reproduced in Fig. 9. Two different domains of jerky flow expand towards each other with increasing strain rates, and merge at a well-defined strain value. The same behaviour is recorded when temperature is changed from 270 to 250°C at a constant strain rate of $6.5 \times 10^{-5} \text{ s}^{-1}$ [27]. This is exactly the behaviour expected from the present model in the high strain rate or low temperature limit, cf. e.g. Fig. 6 (1). In these two examples we notice that (i) the critical strains other than the first one may show either a normal or an inverse behaviour depending on whether they are associated with the ascending or descending branch of the elementary strain profile, and (ii) that these characteristic types of behaviour are only obtained within narrow ranges of strain rates and temperatures, which perhaps explains the scarcity of such experimental data.

4.5. The vacancy model vs the strain hardening model

In classic approaches, the elementary strain is set proportional to the mobile dislocation density, and the latter is assumed to increase as a power β of strain. It follows that in interstitial f.c.c. or b.c.c. alloys, double logarithmic plots of the first critical strain, ϵ_c , vs the strain rate (or semi-logarithmic plots $\ln \epsilon_c$ vs $1/T$) yield a straight line with slope β^{-1} , with β of the order of 0.5–1 [10, 11]. In substitutional alloys, this slope is $(m + \beta)^{-1}$, with $m + \beta$ of the order of 2–3 [9].

According to the vacancy model [9], this difference in the behaviour of interstitial and substitutional alloys stems from the deformation-induced production of excess vacancies whose concentration increases as ϵ^m . A competing approach is the "strain hardening model" of Kocks and co-workers [7, 12, 18]. The model was motivated by the experimental observation that during DSA the strain rate

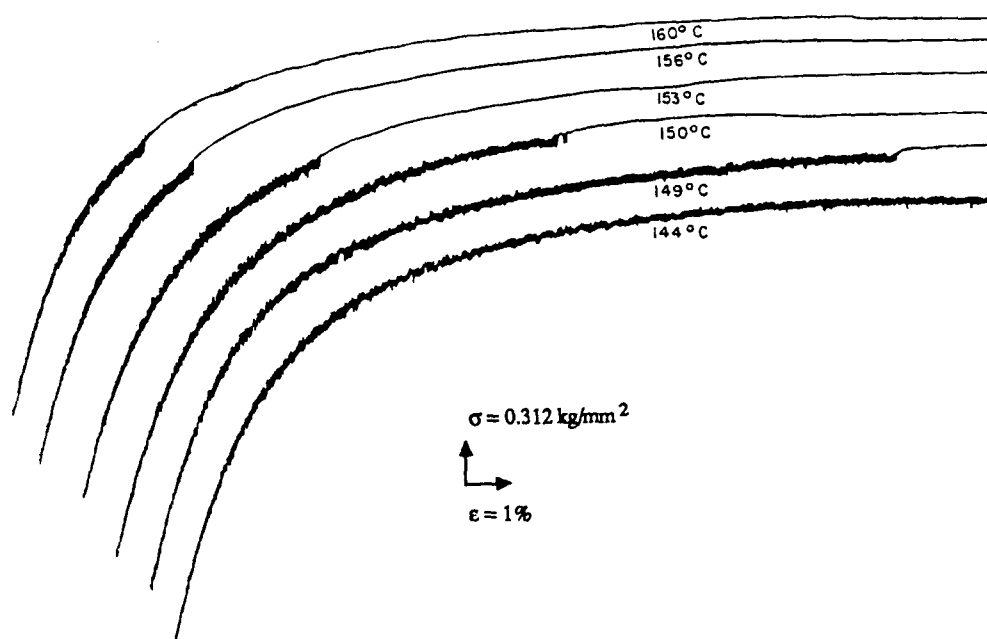


Fig. 8. Disappearance of jerky flow at large strains in an Al-0.5% Mg alloy. The strain rate is $3.33 \times 10^{-4} \text{ s}^{-1}$ and the grain size is 0.08 mm. After Guillot and Grilhé [20].

sensitivity linearly decreases with stress. (At a later stage it increases again, which is attributed to the onset of dynamic recovery.) In a joint paper by van den Beukel and Kocks [4], a juxtaposition of the two models was given.

The present model is rather neutral with respect to the debate of Ref. [4]. It suggests that both competing approaches could be perfected in order to yield a better assessment of experimental results. In some features it is akin to the strain hardening model.

This is seen e.g. from equation (16) of Ref. [4], which expresses the stress dependence of the SRS during DSA

$$S = M\sigma_d - K\sigma_d(D\Omega/\dot{\epsilon})^{2/3}. \quad (7)$$

Here σ_d is the contribution of dislocation-dislocation interactions to the flow stress, and the SRS in the absence of aging, $M\sigma_d$, obeys the Cottrell-Stokes law. The constant K incorporates quantities which are constant during a deformation test and D is the

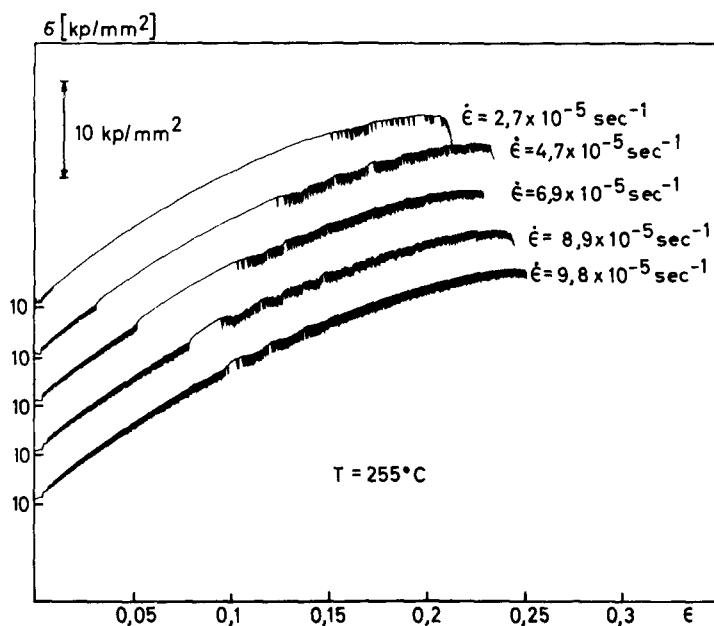


Fig. 9. Deformation curves with two distinct domains of serrated flow in a Cu-3.3 at.% Sn alloy at 255°C (grain size 50 μm). After Räuchle, Vöhringer and Macherauch [27]. (Courtesy Prof. E. Macherauch.)

(constant) effective diffusion coefficient. It is stated in Ref. [4] that the critical strain is a complicated function which cannot be determined easily. We see, however, that when $S = 0$, equation (7) yields $(D\Omega/\dot{\epsilon})^{2/3} \approx X = \text{const.}$, i.e. the same condition as the one used in Section 2, except that there is only one critical value for X since the concentration of solute atoms near dislocations is not assumed to saturate. Taking into account the strain (or stress) dependence of the elementary strain obtained within the present model, the stress dependence of the SRS given by equation (7) exhibits a short strong initial decrease, when the influence of DSA is large enough. This stage is followed by a long quasi-linear decrease, when $\Omega(\epsilon)$ approaches its flat maximum, and by a subsequent increase. This pattern qualitatively fits the experimental observations [7, 18]. However, for particular shapes of the profile $\Omega(\epsilon)$, the quasi-linear stage can be rather short and significant deviations from linearity can occur in the curve S vs σ_a . Therefore, a detailed description of the elementary incremental strain on the basis of the present treatment would definitely enrich the strain hardening model [7, 12, 18].

A certain parallelism between the present approach and the strain hardening model does not rule out the vacancy model, though serious arguments against it have been put forward in the literature [4, 7, 8]. By no means can it be taken for granted that vacancy production during straining prevails over interstitial generation. Neither has a mechanism been identified which could account for a power-law increase of the excess vacancy concentration required by the model [4, 9]. On the other hand, rejecting the vacancy model would cause serious difficulties with explaining the difference in the observed ϵ_c vs $\dot{\epsilon}$ behaviour in interstitial and substitutional alloys. It would be necessary then to prove that some elementary dislocation mechanism (e.g. multiplication) underlying the PLC effect significantly differs in interstitial and substitutional alloys, no matter whether of f.c.c. or b.c.c. structure. It does not appear possible, at this stage, to suggest such a mechanism.

Without feeling a pressure of necessity to make a final choice in favour of a particular model now, we just maintain that a more refined description of the elementary strain suggested provides a better basis for a comparison of the competing models. Possibly, an extended evolutionary model including the vacancy concentration as a third "structural variable" coupled with the mobile and forest dislocation densities will reconcile them and provide an adequate solution to the problem.

5. CONCLUSION

The consideration of the critical strains in the present model was based on the commonly accepted condition of vanishing of the strain rate sensitivity. A new step consists in a more refined treatment of the elementary strain Ω that incorporates the micro-

structural features on the dislocation level. By considering the evolution of two coupled dislocation densities, it was shown that the whole variety of experimentally observed situations with regard to the strain ranges of existence of the PLC effect can be accounted for. The considerations outlined in the present paper are neutral to the particular choice of the DSA model and can be implemented in a more general approach. A more detailed description of the evolution of the dislocation densities during straining would certainly be helpful in discriminating between competing models for dynamic strain aging.

Acknowledgements—The authors would like to express their gratitude to DAAD and ANRT for financial support through the PROCOPE program. The assistance of Dr A. Styczynski with numerical work is gratefully appreciated.

REFERENCES

1. A. van den Beukel, *Physica status solidi* (a) **30**, 197 (1975).
2. U. F. Kocks, in *Progress in Materials Science*, Chalmers Anniversary Volume. Pergamon Press, Oxford (1981).
3. P. G. McCormick, *Acta metall.* **36**, 3061 (1988).
4. A. van den Beukel and U. F. Kocks, *Acta metall.* **30**, 1027 (1982).
5. A. H. Cottrell, *Phil. Mag.* **44**, 829 (1953).
6. L. J. Cuddy and W. C. Leslie, *Acta metall.* **20**, 1157 (1972).
7. R. A. Mulford and U. F. Kocks, *Acta metall.* **27**, 1125 (1979).
8. R. B. Schwarz and L. L. Funk, *Acta metall.* **33**, 295 (1985).
9. A. van den Beukel, *Acta metall.* **28**, 965 (1980).
10. Y. Nakada and A. S. Keh, *Acta metall.* **18**, 437 (1970).
11. P. G. McCormick, *Acta metall.* **21**, 873 (1973).
12. U. F. Kocks, R. E. Cook and R. A. Mulford, *Acta metall.* **33**, 623 (1985).
13. Y. Estrin and L. P. Kubin, *Acta metall.* **34**, 2455 (1986).
14. U. F. Kocks, A. S. Argon and M. F. Ashby, *Thermodynamics and Kinetics of Slip*, *Progr. Mater. Sci.*, Vol. 19. Pergamon Press, Oxford (1975).
15. L. P. Kubin, K. Chihab and Y. Estrin, *Acta metall.* **36**, 2707 (1988).
16. N. Louat, *Scripta metall.* **15**, 1167 (1981).
17. J. Friedel, *Dislocations*. Pergamon Press, Oxford (1964).
18. P. Wycliffe, U. F. Kocks and J. D. Embury, *Scripta metall.* **14**, 1349 (1980).
19. S. Miura, in *Mechanical Behaviour of Materials*, Vol. 1, p. 128. Soc. Mater. Sci., Japan (1972).
20. J. Guillot and J. Grilhé, *Acta metall.* **20**, 291 (1972).
21. P. G. McCormick, *Scripta metall.* **6**, 165 (1972).
22. E. Pink, in *Strength of Metals and Alloys (ICSMA 8)* (edited by P. O. Kettunen, T. K. Lepistö and M. E. Lehtonen), Vol. 1, p. 494. Pergamon Press, Oxford (1988).
23. E. Pink and A. Grinberg, *Mater. Sci. Engng.* **51**, 1 (1981).
24. R. W. Hayes and W. C. Hayes, *Acta metall.* **32**, 259 (1984).
25. K. Mukherjee, C. D'Antonio, R. Macia and G. Fisher, *J. appl. Phys.* **39**, 5434 (1968).
26. D. Munz and E. Macherauch, *Z. Metallk.* **57**, 553 (1966).
27. W. Räuchle, O. Vöhringer and E. Macherauch, *Mater. Sci. Engng* **12**, 147 (1973).

APPENDIX

The Strain Dependence of the Dislocation Densities

The following set of two coupled differential equations is used to describe the evolution with strain of the densities of mobile and forest dislocations

$$d\rho_m/d\epsilon = C_1/b^2 - C_2\rho_m - (C_3/b)\rho_f^{1/2} \quad (\text{A1})$$

$$d\rho_f/d\epsilon = C_2\rho_m + (C_3/b)\rho_f^{1/2} - C_4\rho_f. \quad (\text{A2})$$

Here the parameters C_i are related to the multiplication of mobile dislocations (C_1), their mutual annihilation and trapping (C_2), their immobilization through interaction with forest dislocations (C_3), and to dynamic recovery (C_4). The parameter C_2 is reintroduced in the equation for the evolution of the forest density to account for the creation of immobile dipolar arrangements and for the formation of debris, jogs, loops, left by the mutual annihilation of mobile dislocations. In what follows these quantities will be considered as material constants. It should be kept in mind, however, that some of them, especially C_2 and C_4 , may depend on strain rate and temperature since they account for thermally activated mechanisms like cross-slip or climb. The system of equations (A1) and (A2) is similar to the one discussed in a previous work [13] except that a different type of multiplication mechanism is assumed here for the sake of simplicity. Not taking into account the loss terms in equation (A1), the mobile density would increase linearly with strain. The rate of increase with time, $(d\rho_m/dt)^+$, which can be deduced from the rate of increase with strain with the help of Orowan's law, is proportional to ρ_m and to the average dislocation velocity, while the first term in the rate of density loss $(d\rho_m/dt)^-$ is proportional to $-\rho_m^2$, in conformity with many phenomenological elaborations. Taking into account the loss terms, the increase of ρ_m may be represented as a power of strain slightly smaller than 1, to be compared with values found in the literature on substitutional alloys [9].

The solution of equations (A1) and (A2) shows that the two densities tend towards saturation, the mobile density saturating faster than the forest density, cf. Ref. [13]. The solution is not stable at small strains, where deformation is always very heterogeneous and, for this reason, strain values much below 10^{-3} are not considered. Finally, it was checked numerically, using the same procedure as the one described in the next section, that introducing other possible types of multiplication mechanisms [14] does not substantially modify the conclusions drawn below.

Numerical Estimates

To obtain typical estimates for the strain dependences of ρ_m , ρ_f and Ω , one has to define the initial values of the dislocation densities and to specify four conditions determining a reference set of numerical values for the material constants. The particular choice here is as follows.

—The initial dislocation densities are fixed to the values $\rho_m = 10^{10} \text{ m}^{-2}$, $\rho_f = 10^{11} \text{ m}^{-2}$, so that with $b = 3 \times 10^{-10} \text{ m}$ the elementary strain has an initial value of $\Omega_0 = 9.5 \times 10^{-6}$.

—The material constants are determined through the following conditions. (i) The saturated densities are taken to be $\rho_{ms} = 5 \times 10^{13} \text{ m}^{-2}$ and $\rho_{fs} = 10^{14} \text{ m}^{-2}$ for the mobile and forest dislocations respectively, so that the saturation value of the elementary strain is $\Omega_s = 1.5 \times 10^{-3}$. Though set arbitrarily, these values of ρ_{ms} and ρ_{fs} are fairly representative. It should be noted, however, that these quantities are

generally strain rate dependent. This dependence will be disregarded for simplicity, however. (ii) The relaxation strain associated with the forest density, i.e. the strain ϵ_f at which $\rho_{fs} - \rho_f = (\rho_{fs} - \rho_{\bar{f}})/e$, is fixed to a value $\epsilon_f = 0.3$, again disregarding an ϵ dependence. (iii) In equation (A1), two mechanisms contribute to the saturation of the mobile density. Their relative weight at saturation is measured through a constant $\lambda = C_2\rho_{ms}/[(C_3/b)\rho_{fs}^{1/2}]$, which depends on the material considered as well as on the experimental conditions. Assuming that immobilization of the mobiles on the forest is more efficient than their annihilation or trapping, a reasonable value would be $\lambda = 0.1$. A larger value would indicate, for instance, a greater cross-slip ability of screw mobiles.

At saturation, $d\rho_m/d\epsilon = d\rho_f/d\epsilon = 0$. In such conditions, equations (A1) and (A2) yield

$$\rho_{fs} = C_1/(b^2 C_4) \quad (\text{A3})$$

and

$$C_2\rho_{ms} = (C_1/b^2) - (C_3/b)\rho_{fs}^{1/2}. \quad (\text{A4})$$

At large strains, the mobile density is close to saturation and we assume that $d\rho_m/d\epsilon \approx 0$. Then, ρ_m can be determined from equation (A1) and substituted into equation (A2) leading to

$$d\rho_f/d\epsilon = C_4(\rho_{fs} - \rho_f). \quad (\text{A5})$$

The forest density saturates with strain obeying an exponential form $\rho_{fs} - \rho_f = (\rho_{fs} - \rho_{\bar{f}}) \exp(-C_4\epsilon)$, and condition (ii) gives

$$C_4 = 1/\epsilon_f. \quad (\text{A6})$$

Finally, condition (iii) is expressed as follows

$$\lambda = C_2\rho_{ms}/[(C_3/b)\rho_{fs}^{1/2}] = C_2\Omega_s/C_3. \quad (\text{A7})$$

Combining equations (A3), (A4), (A5) and (A6), we obtain

$$C_4 = C_1/(b^2\rho_{fs}) = 1/\epsilon_f, \quad (\text{A8})$$

and

$$C_3/b = (C_2/b)\Omega_s/\lambda = \rho_{fs}^{1/2}/[\epsilon_f(1 + \lambda)]. \quad (\text{A9})$$

As a result, we obtain the following set of constants for $\lambda = 0.1$: $C_1/b^2 = 10^{15}/3$, $C_2 = 6.06$, $C_3/b = 10^8/3.3$, $C_4 = 10/3$. Fig. 2 of the text shows the strain dependence of the mobile and forest densities obtained by solving numerically the system of equations (A1) and (A2) for such conditions.

The Elementary Strain

The calculated strain dependence of the elementary strain Ω is given in curve (a) of Fig. 3 (see text). As expected, this diagram exhibits a maximum, at a strain of about 16.5%. To check the influence of the parameter values on the occurrence and position of the maximum, the material constants were modified one by one with respect to the basic set of numerical values defined above. Other curves of Fig. 3 were obtained in the following conditions: $\lambda = 1$ (b), $\epsilon_f = 0.5$ (c), $\rho_{fs} = 10^{12} \text{ m}^{-2}$ (d), the rest of the parameter values being unchanged. The effect of the initial value of the mobile dislocation density on the shape of the Ω vs ϵ profile was shown to be less significant. In all cases a maximum was obtained for strain values ranging from a few percent to about 30%. A decrease of the saturated mobile density ρ_{ms} to $5 \times 10^{12} \text{ m}^{-2}$ induces a very strong increase of the maximum which is shifted towards small strains. The only way to suppress the maximum is to reduce ρ_{fs} to sufficiently low values (e.g. 10^{13} m^{-2}).